

Anton Roupasov-Ruiz

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EDUCATION

University of Alberta

Edmonton, Alberta, Canada

4th year BSc Honors in Computing Science, cGPA: 3.54 / 4.00

Sep 2023 – expected Jun 2027

- **Related Coursework:** Reinforcement Learning, Machine Learning, Search and Planning in AI, Program Synthesis, Heuristic Search and Artificial Life Research Seminar, Software Engineering Principles, Data Structures & Algorithms, Design & Analysis of Algorithms, Computing in Python, Computing in C & C++, Calculus III, Linear Algebra, Discrete Mathematics, Applied Statistics

INDUSTRY EXPERIENCE

Microsoft

Redmond, Washington, United States

Software Engineer Intern

May 2026 – Present

- Collaborated with PMs and engineers over a **12-week timeline** to scope project requirements, successfully breaking down ambiguous goals into actionable technical milestones and architecture designs.
- Engineered a production-grade **AI Agent** integrated with **Model Context Protocol (MCP)** tools for a new internal platform, enabling users to translate **natural language** to dynamically generate, debug, and deploy complex enterprise agent workflows.
- Architected the backend to orchestrate model invocation via **Azure AI Foundry** while utilizing **Azure Cosmos DB** to manage and persist agent conversation histories.
- Accelerated development cycles by leveraging **GitHub Copilot CLI** to manage distributed **agent fleets** and autonomous **subagents**.
- Engaged in regular Scrum cycles and delivered continuous feature demos to engineers and PMs, driving rapid feedback loops to iterate on user experience and product alignment.

Microsoft

Redmond, Washington, United States

Explore (Software Engineering and Product Management) Intern

May 2025 – Aug 2025

- Collaborated cross-functionally with PMs and engineers to identify **Bing ETL pipeline bottlenecks** and evaluate tradeoffs of architectural options, ultimately outlining and implementing the optimal approach.
- Developed and pushed **production-ready fullstack project** with a **React (TypeScript) frontend** and a **Java backend**, reducing **ETL debugging latency** from **15 minutes to seconds** by delivering a **containerized, scalable solution** with **Azure Kubernetes Service, Azure Functions, and Azure SignalR Service**, integrating robust **API design and web protocols** for reliable, low-latency communication across distributed systems.
- Authored clear **technical documentation** and implemented **comprehensive test suites**, leveraging **GitHub Copilot Agent Mode** to accelerate development while maintaining quality.
- Presented internship project live demo and quantified impact to stakeholders across Microsoft Bing, including **CVPs and VPs**.

RESEARCH EXPERIENCE

University of Alberta

Edmonton, Alberta, Canada

Undergraduate Researcher

Sep 2025 – Dec 2025

- Conducted a **semester-long research project** in **CMPUT 497/651 (graduate-level artificial intelligence research seminar focused on program synthesis, heuristic search, and artificial life)** under **Dr. Vadim Bulitko (Reinforcement Learning and Artificial Intelligence Lab, University of Alberta)**, developing the first fully specified, open-source reimplementation of the classic Ackley & Littman (1991) predator-prey **evolutionary reinforcement learning A-Life testbed**. [\[arXiv link\]](#)
- Formulated explainable, modular **soft, differentiable decision lists** as a programmatic alternative to black-box neural networks, evaluating policy viability across **4,000 independent trials** ($\tau_{trial} = 2000$) using Kaplan-Meier curves and restricted mean survival time metrics absent in the original study.
- Proved programmatic agents significantly outlasted neural networks ($\log\text{-rank } p \approx 4.1 \times 10^{-72}$), surviving an average of **201.69 steps longer** (95% Confidence Interval [180.14, 223.25], $Z = 18.34$, $p \approx 3.8 \times 10^{-75}$); demonstrated that learning-only decision lists (no evolution) outlived evolved neural networks by **73.67 steps** ($p \approx 4.7 \times 10^{-12}$).

- Delivered **weekly AI literature synopses** on cutting-edge AI research papers related to **program synthesis, heuristic search, and artificial life**.
- Presented a **60-minute individual seminar** on “**Attention Is All You Need**” to the class, facilitating technical discussions on transformer architecture evolution.

University of Alberta

Edmonton, Alberta, Canada

Undergraduate Researcher

May 2024 – Aug 2024

- Conducted **literature review** and authored **4000+ word analysis** of **4 computational associative memory models** under the supervision of **Dr. Jeremy Caplan (Computational Psychology Lab, University of Alberta)** while a **first-year student** as part of a **3rd-year Honors Psychology independent research course**, achieving a grade of **4.00/4.00** as a result. [\[Full paper\]](#)
- Implemented MATLAB simulations of **neural-network, convolution-based, local trace, and matrix models**, producing reusable code and documentation that improves onboarding resources for new lab members.

PROJECTS

Prelude (Microsoft Global Intern Hackathon)

Redmond, Washington, United States

Software Engineer Intern

Jul 2026

- Co-engineered an innovative framework within a team of 6 to automatically generate **Model Context Protocol (MCP) tools** for apps in the Microsoft Store ecosystem by programmatically crawling the OS accessibility tree.
- Presented the architecture and live product capabilities at the global intern hackathon science fair, demoing the working system to engineering leadership, product managers, and team members.

MentAI

Edmonton, Alberta, Canada

Software Developer

Dec 2024 – Apr 2025

- Engineered a Swift-based iOS MVP utilizing **Cursor** and the **OpenAI ChatGPT API** to explore LLM-assisted wellness insights and real-time personalized guidance. [\[MVP Video\]](#)
- Pitched the MVP to the **Alberta Machine Intelligence Institute (Amii)**—one of Canada’s three federally funded AI research institutes—and the **University of Alberta Health Sciences Startup Incubator**.

EEG-Based Pomodoro Focus Timer

Edmonton, Alberta, Canada

Software Developer

Nov 2024

- Built a **Python-based** sensor-driven Pomodoro focus timer to **monitor concentration levels during study sessions** integrating **EEG data streams** and **signal processing** with the **Python BrainFlow** library for the **natHacks 2024 neurotech hackathon**. [\[GitHub\]](#)

Lamda Studios

Edmonton, Alberta, Canada

Software Developer

Nov 2023 – Feb 2024

- Built and launched **3 interactive games** using the **Unreal Engine Editor for Fortnite** that attracted **71+ concurrent players** and **650+ gameplay hours within two weeks**, demonstrating ability to deliver engaging, scalable experiences. [\[Showcase\]](#)
- Executed a multi-platform marketing campaign with 30,000+ cross-platform views, reflecting ability to **scale systems and communicate impact to wide audiences**. [\[Showcase\]](#)

LEADERSHIP

University of Alberta Undergraduate AI Society

Edmonton, Alberta, Canada

Panelist, “AI in Industry”

Sep 2025

- Shared insights on industry SWE experience at Microsoft

Microsoft – VP Level Monthly Diversity & Inclusion Event

Redmond, Washington, United States

Volunteer Speaker

Jul 2025

- Delivered a talk to **100+ employees across Bing, including all teams under my VP**, sharing cultural celebrations from my heritage.

neurAlbertaTech – University of Alberta Chapter

Edmonton, Alberta, Canada

Treasurer

Sep 2024 – May 2025

- Oversaw finances and annual budget for a **student-led neurotechnology club**.
- Recruited new members at campus outreach events.

Dean’s Honor Roll / First Class Standing

University of Alberta

Apr 2026, Apr 2025

Jason Lang Scholarship

University of Alberta

Nov 2025, Nov 2024

Alexander Rutherford Scholarship

Alberta Student Aid

Sep 2023

AP Scholar with Distinction

College Board

Jun 2023

AP Scholar Award

College Board

Jul 2022

SKILLS

Programming Languages: C#, TypeScript, Python, JavaScript, Java, C++, C, MATLAB, HTML/CSS, LaTeX
Tools: React, Azure Kubernetes Service, Azure Functions, Azure SignalR Service, GitHub Copilot, Git, Bash, Linux, Ubuntu, Cursor, VSCode, Flask, Unreal Engine Editor
Databases: Google Firebase (NoSQL)
ML & Data: RL-glue, Scikit-learn, Pandas, NumPy, Matplotlib
Languages: English (native), Spanish (native), Russian (native), French (intermediate)